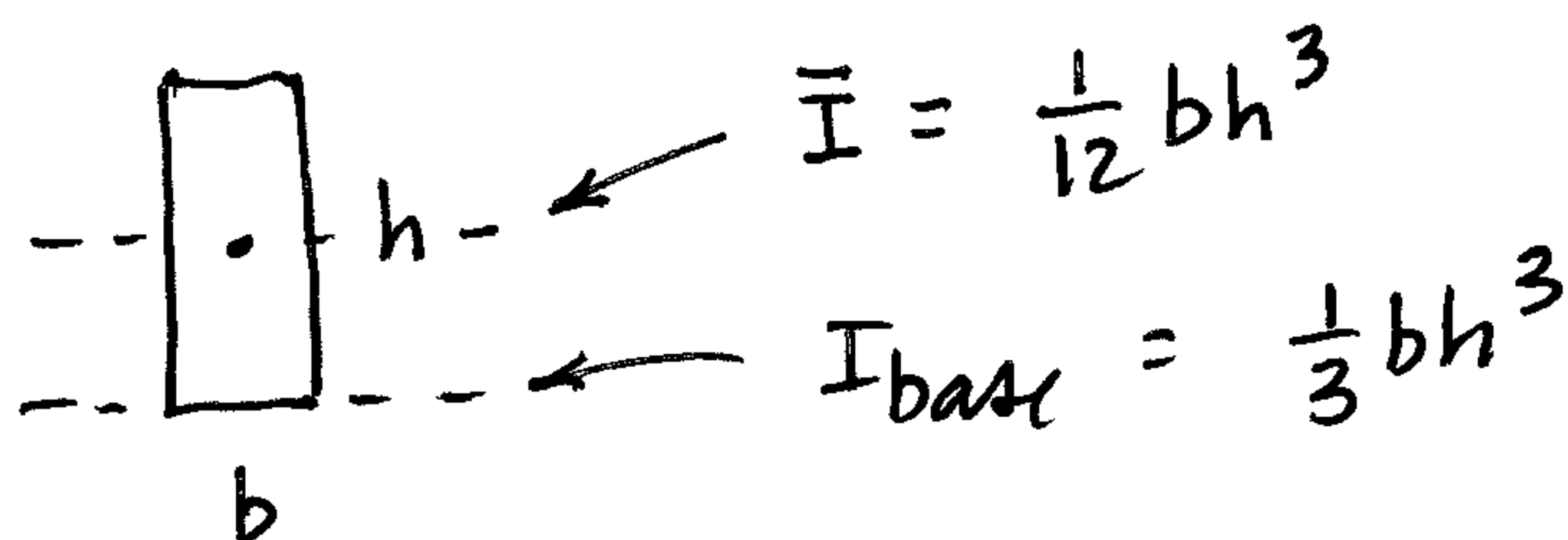


# Area Moment of Inertia (MOI) Formulas

For each shape, I prefer to give only one set of MOI formulas, e.g.

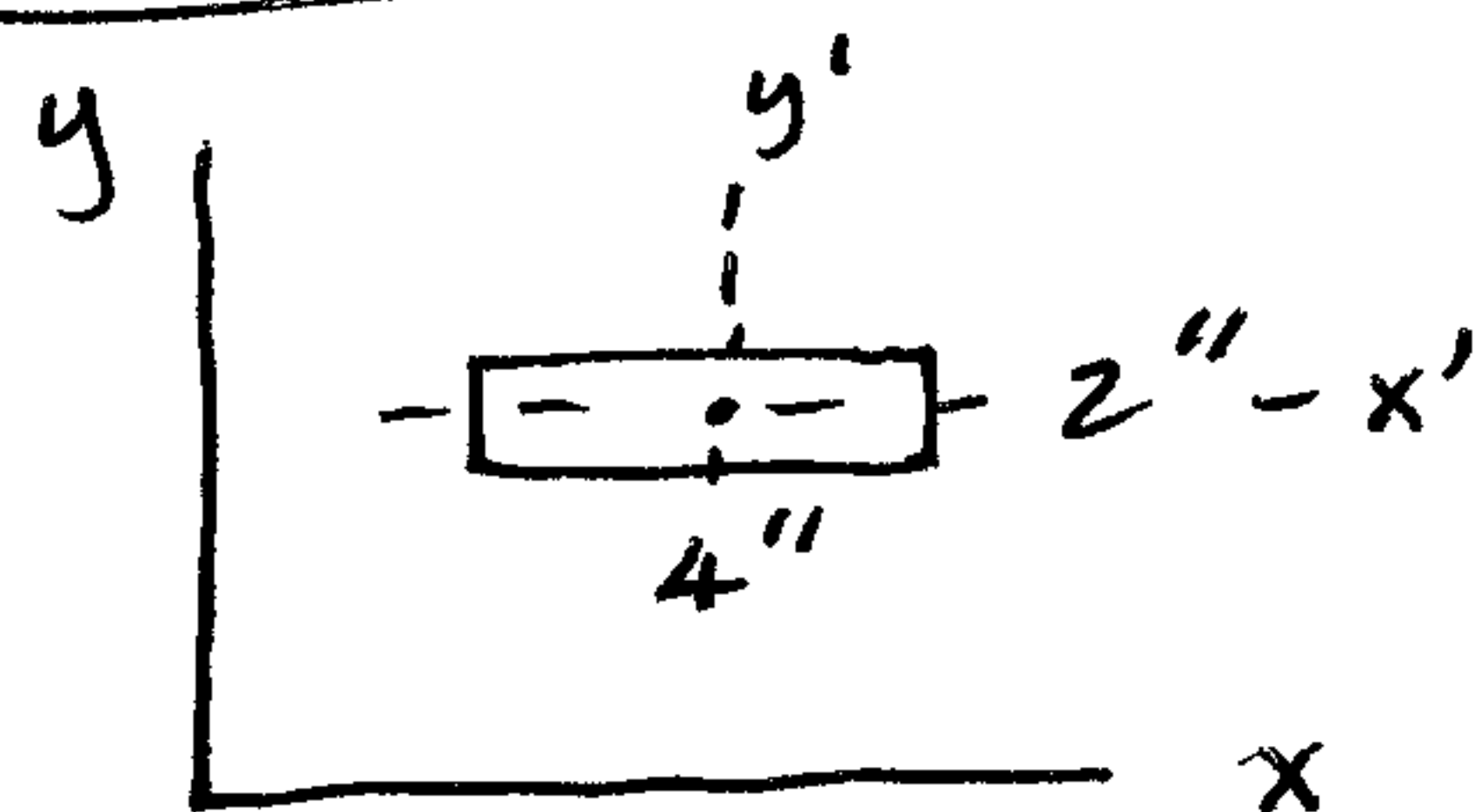


To use the formula, the user identifies the dimensions of  $b$  and  $h$ .

"Base" -- the dimension that is parallel to the axis of interest

"Height" -- the dimension that is  $\perp$  to the axis of interest

Examples:

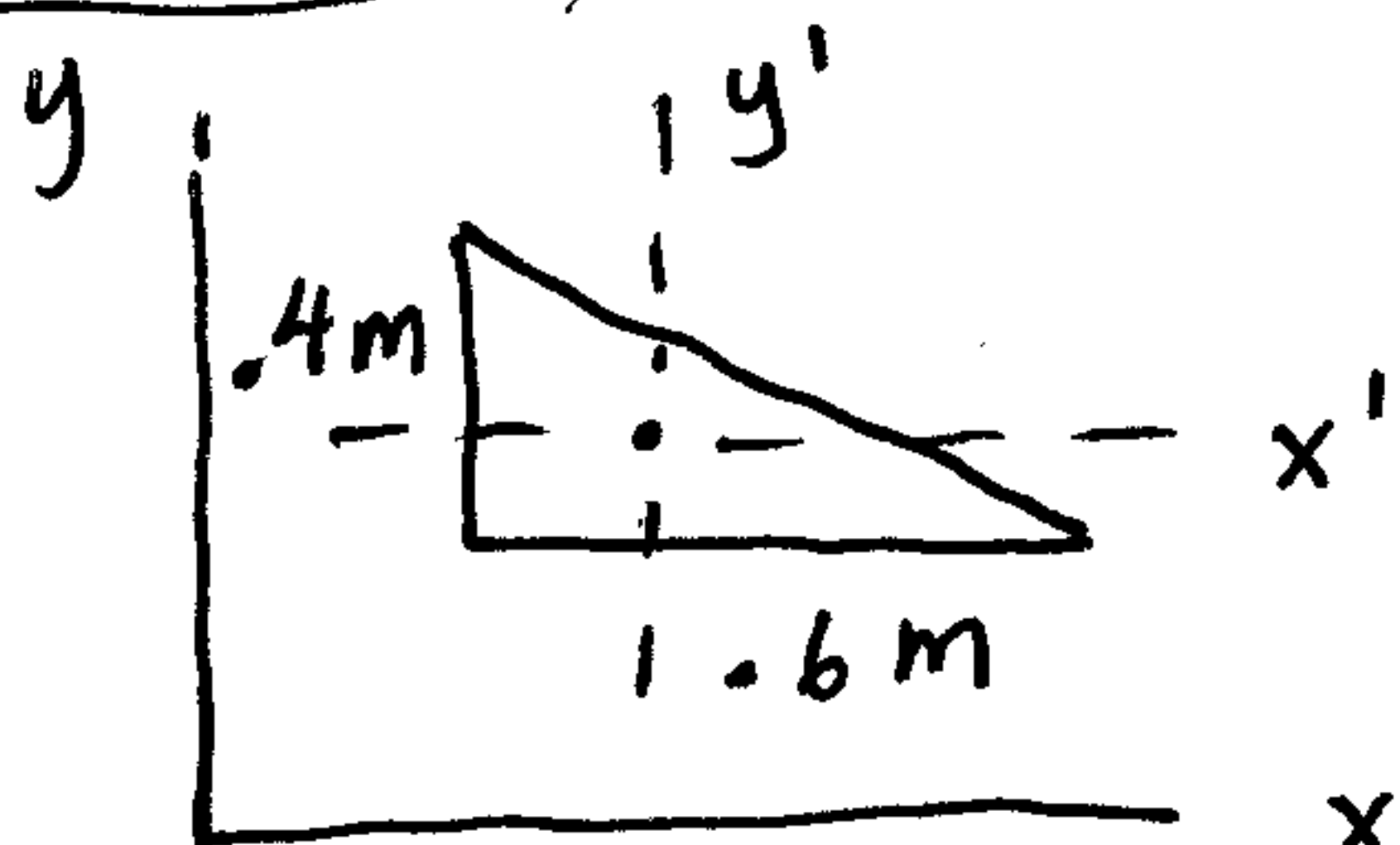


wrt  $x$ ,  $4'' = \text{base}$   
 $2'' = \text{height}$

$$\bar{I}_{x'} = \frac{1}{12}bh^3$$
$$= \frac{1}{12}(4)(2)^3$$

wrt  $y$ ,  
 $2'' = \text{base}$   
 $4'' = \text{height}$

$$\left\{ \text{and } \bar{I}_{y'} = \frac{1}{12}bh^3 = \frac{1}{12}(2)(4)^3 \right.$$



$$\boxed{\bar{I} = \frac{1}{36}bh^3}$$

wrt  $x$ ,  $0.6\text{ m} = b$ ,  $0.4\text{ m} = h$

$$\text{thus, } \bar{I}_{x'} = \frac{1}{36}(.6)(.4)^3$$

wrt  $y$ ,  $0.4\text{ m} = b$ ,  $0.6\text{ m} = h$

$$\text{thus, } \bar{I}_{y'} = \frac{1}{36}(.4)(.6)^3$$

See the Hibbeler crib sheet: He gives two formulas for each shape... and he reversed the  $b$  and  $h$ .